

SENTHIL KUMAR NAGAPPAN

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AI Safety & Human Systems Integration Leader | Product Design | Enterprise Governance | Regulated Systems

U.S. Citizen | Active Public Trust Clearance | TS/SCI Eligible

EXECUTIVE SUMMARY

Senior AI and Human Systems Integration leader with 18+ years building enterprise functions that deliver safe, compliant, AI-driven products across healthcare, federal, defense, financial services, and consumer electronics environments. Rare combination of three high-demand disciplines in one professional: AI safety and governance, human factors engineering, and enterprise accessibility.

Created **SSA11y** (Llama2) — AI-powered enterprise platform now the federal compliance standard (\$1.5M+ annual savings). Established AI safety governance at GE HealthCare — zero AI features ship without passing this framework. Directed \$130M+ portfolios impacting 50M+ users. Expert in Figma, design systems, enterprise design governance, user research, and design-to-engineering collaboration in Agile environments.

ENTERPRISE IMPACT & LEADERSHIP OUTCOMES

Enterprise AI Governance & Human-AI Interaction Standards

Established AI safety framework and human-in-the-loop decision models for clinical AI at GE HealthCare — defining evaluation criteria, deployment thresholds, trust and explainability standards, and human-AI collaboration patterns (supervision, delegation, escalation). Adopted as enterprise standard across 10+ teams, 15+ leaders, 50+ stakeholders. Zero AI-enabled features now ship without passing this framework.

AI-Powered Product: Shipped, Adopted, Scaled

Conceived, built, and deployed SSA11y (Llama2) — GenAI-powered platform with agentic workflows, prompt engineering, and automated validation integrated into CI/CD pipelines across 40+ monthly releases. Reduced audit effort 30%, saved \$1.5M+ annually, eliminated 200+ violations. Now the operational standard across federal systems.

Wearable & Consumer Product Safety (MIL-STD-1472H)

Applied DoD human engineering design criteria to consumer healthcare wearable devices at Best Buy Health — redesigning emergency activation controls for 1M+ elderly users. Conducted usability studies with users aged 65–90+ under simulated stress. Eliminated critical interaction failures. Governance adopted across 5 product lines.

\$130M Federal Portfolio Leadership (7 Years)

Directed \$130M federal healthcare portfolio at SAMHSA end-to-end — budget, staffing, vendor accountability, design governance, and investment strategy across 6+ platforms serving 2M+ monthly visitors. Reduced rework 30–40%, saved \$2M+ annually. Defined Section 508 as organizational policy.

CORE EXPERTISE

AI & GenAI AI safety, GenAI/LLM (Llama2), human-in-the-loop systems, conversational AI, AI copilots, agentic AI, prompt engineering, trust and explainability, hallucination controls, ChatGPT, Claude AI, Copilot, Cursor

Human Systems Integration HSI (DoDI 5000.95), MIL-STD-1472H, human factors engineering, ergonomics, cognitive and physical ergonomics, anthropometry, human error analysis, human-AI teaming, qualitative and quantitative research

Product Design & UX End-to-end UX design, interaction design, information architecture, wireframes, prototyping, user research, usability testing, design systems, Figma, FigJam, Miro, Adobe CC, JAWS, NVDA, VoiceOver

Regulatory & Compliance FDA Human Factors, IEC 62366, ISO 14971, FMEA, URR, WCAG 2.2 AA, Section 508, ADA Title II, MBSE, SysML 2, medical device usability, patient safety

Technical React, React Native, TypeScript, HTML, CSS, JavaScript, Tailwind CSS, Salesforce Lightning (SLDS), USWDS, CI/CD, Agile/Scrum, Rally, Jira, Aha!, Confluence

Leadership Enterprise governance, portfolio ownership (\$130M+), cross-functional leadership (10+ teams), executive advisory, investment strategy, vendor management, stakeholder alignment (50+)

PROFESSIONAL EXPERIENCE

AI + Human Systems Integration Lead | Brain Pulse Inc, GE HealthCare

Mar 2025 – Present

Enterprise AI Safety & Clinical Systems | FDA, IEC 62366, ISO 14971, MIL-STD-1472H

Established the AI safety function and enterprise operating model, accelerating regulatory approvals, product delivery, and operational efficiency across the organization.

- **Own and lead AI safety and HSI function across the enterprise**, including org design, hiring, and performance management. Defined human-in-the-loop decision frameworks, AI trust and explainability standards, and deployment governance.
- **Lead cross-functional organization of 10+ teams** across engineering, product, clinical, regulatory, and QA. Partner with senior leadership to shape AI product strategy, roadmap, and design direction.
- **Designed AI-powered interfaces and enterprise design system in Figma** — creating reusable components, interaction patterns, and visual standards. Worked closely with engineers from inception through implementation in Agile.
- **Institutionalized AI safety governance as enterprise standard** adopted across 10+ product teams, 15+ leaders, 50+ stakeholders. Accelerated FDA submission readiness and reduced late-stage design rework.

Sr. UX & Accessibility Lead | Leidos Inc, Social Security Administration Oct 2022 – Feb 2025
Federal Enterprise Platform (15+ modules, 50M+ users) | Public Trust | WCAG 2.2 AA, Section 508
Led enterprise UX, accessibility, and AI integration strategy for a national-scale federal platform impacting 50M+ citizens.

- **Conceived, built, and deployed SSA11y** — AI-powered platform (Llama2) with GenAI interfaces, prompt-driven workflows, and automated validation. Adopted across 40+ monthly releases. Reduced audit effort 30%, saved \$1.5M+ annually.
- **Defined enterprise design system and accessibility architecture** across 15+ product modules in Figma — reusable components, interaction patterns, and governance. Eliminated 200+ violations, reduced defects 30–40%.

Inclusive Design & Accessibility Lead | Best Buy Health Sep 2021 – Sep 2022
Healthcare Wearables (5 product lines, 1M+ users) | WCAG 2.2 AA, FDA, IEC 62366, MIL-STD-1472H
Defined accessibility governance and wearable product safety standards, improving outcomes for 1M+ elderly users.

- **Applied MIL-STD-1472H to redesign wearable emergency controls** — addressing degraded dexterity, vision, and cognitive load under stress. Conducted usability studies with users aged 65–90+. Eliminated missed emergency activations.
- **Established design governance across 5 product lines** with enterprise design system, accessible components, and cross-platform UX (React Native + web). Reduced remediation cycles 30–40%.

Sr. UX & Accessibility Program Manager | Eagle Technologies Inc, HHS/SAMHSA Mar 2014 – May 2021
\$130M Federal Portfolio (6+ platforms, 15+ team, 50+ stakeholders) | Section 508, WCAG 2.0 AA
Directed \$130M federal portfolio end-to-end, institutionalizing design governance and influencing executive funding strategy.

- **Owned and directed \$130M federal portfolio** across 6+ enterprise platforms serving 2M+ monthly visitors — UX, accessibility, and engineering delivery with full budget, staffing, and vendor accountability.
- **Defined investment prioritization and design governance** across 50+ stakeholders. Institutionalized HSI-driven design, reducing rework 30–40%, saving \$2M+ annually. Defined Section 508 as organizational policy.

EARLIER EXPERIENCE

Lead UX Designer — AAMC | **Sr. UX & Accessibility Consultant** — CareFirst BCBS (Healthcare Insurance)
Accessibility Analyst — JPMorgan Chase (FinTech/Banking) | **Human Factors Engineer** — Threewise Corp
UX Consultant (Salesforce Lightning) — Morgan Stanley (Wealth Management)

EDUCATION & CERTIFICATIONS

Bachelor of Engineering — Electronics & Communication, Madurai Kamaraj University, 1999
CPACC + WAS (IAAP) | **CUA** (HFI) | **CSM** (Scrum Alliance) | **DAU HSI** — In Progress
Recognized Practitioner: W3C WAI, IAAP, HFES

VOLUNTEER LEADERSHIP

Vice President | Texas Astronomical Society 2025 – Present
• Lead organizational strategy, speaker programming, and volunteer coordination for 500+ member non-profit.

Cub Master (Pack 178) | Asst. Scout Master (Troop 187 & 406) | Merit Badge Counselor | BSA 2023 – Present
• Mentor 70+ Cub Scouts and 80+ Boy Scouts in STEM and leadership; manage 12 den leaders.

Lead Coach | FIRST Lego League 2018 – Present
• Coach youth robotics teams through engineering design process.

Affiliations: IAAP, HFES, UXPA, UXPA DC, NOVAC, TAS | W3C WAI (Volunteer)